



TECHNOLOGY

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The PlayStation powered super-computer

Playing the PlayStation and getting your homework done are not usually activities which go hand-in-hand, but one university professor may have cracked it.

Dr Gaurav Khanna has no less than 16 Sony PlayStation 3 consoles working together to create a super-computer which he says rivals the power of more than 200 conventional PCs.

But Dr Khanna is no hardcore gamer. Instead, the University of Massachusetts professor is using the entertainment machines for highly complex calculations.

His research is attempting to go back 13 billion years to learn about the origins of the Big Bang, as well as looking at the black holes that exist in space today.

He explained to Digital Planet just how he manages to harness the consoles to carry out the work.

"The PlayStation comes loaded with GameOS - which is what you use to navigate through to look at pictures and movies and to launch games. But it also has the option, if you dig into the settings, to install [an] alternative operating system."

He installed the open-source operating system Linux onto the machines, before programming software which combined all the consoles to carry out calculations in "parallel" with each other.

"They talk to each other at very high-speed," he said.

"It's set up as a very traditional super-computer. The calculation is broken up in parallel pieces - each PlayStation does its own piece and talks to each other on the fly so that the whole calculation can go forward."

While using a machine designed for the likes of Grand Theft Auto and Call of Duty to test scientific theories may seem strange, Dr Khanna says it is not really that far removed from its intended purpose.

"If you think about it - what is gaming? Ultimately what you're seeing is very beautiful graphics and a lot of movement and so on and so forth, but it's all driven by calculations in the end. I could use the same number-crunching capability to actually solve a hard mathematical problem or hard scientific problem."

The harder task for Mr Khanna was in fact convincing the relevant bodies to give him money to buy the machines.

"There was no way I can go to a federal grant agency or to my university and say 'hey I actually want to buy 16 PlayStations, would you please fund that?'

"So I actually directly talked to Sony. They gave me the first eight that I set up and then later on, once I had quite a bit of success with them, my university helped me purchase the additional eight."

Perhaps ironically, anyone wanting to use the PlayStations in the same way now will be out of luck - Sony has since deactivated the option to install a separate operating system.

"A few months ago, Sony decided to take away the Linux functionality. There are a lot of rumours on why that happened - the most talked about is that somebody, some hacker, used the Linux functionality to actually hack into the PlayStation and run pirate games.

"So that of course caused a panic and Sony has essentially removed the functionality. So what that means basically is that the people who had Linux on the PlayStation are ok, but there's no way of expanding their cluster."

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